

Route C: A Spectral Approach to Zero-Density Estimates, Conditional on Two Analytic Gaps

Corrected: false Lemma A killed; Gap A reframed

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Disclaimer. Conditional only; two analytic gaps remain open. We do *not* claim RH. Archived drafts that evaluated $R(v_{\text{alt}}) \rightarrow -1/(2\pi)$ are **refuted** by numerics.

Item	Status
Lemma A ($\mu(d)\varphi(d)/d^2$ decomposition of Q_N)	False
$R(v_{\text{alt}}) \rightarrow -1/(2\pi)$	False
Parity split of $v_{\text{alt}}^\top Q_N v_{\text{alt}}$	Proved
$\lambda_{\min}(Q_N)/\log N \rightarrow -1/(2\pi)$	Open
Gap B: uniform $\lambda_2 - \lambda_{\min} \geq c_0$	Open

Abstract

Let $Q_N[i, j] = 1/(\gcd(i, j)\sqrt{ij})$ and $v_{\text{alt}}[k] = (-1)^{k+1}/\sqrt{k}$. Earlier Route C drafts claimed $R(v_{\text{alt}}) \rightarrow -1/(2\pi)$ via a Möbius identity that fails off-diagonal. We kill that path and reframe the architecture: numerics support $\lambda_{\min}(Q_N)/\log N \rightarrow -1/(2\pi)$ while $R(v_{\text{alt}})$ diverges from the target. We prove a parity split of the alternating quadratic form and record two honest analytic gaps. No RH claim.

1 Operator and trial vectors

$$Q_N[i, j] = \frac{1}{\gcd(i, j)\sqrt{ij}}, \quad v_{\text{alt}}[k] = \frac{(-1)^{k+1}}{\sqrt{k}}.$$

Then

$$v_{\text{alt}}^\top Q_N v_{\text{alt}} = \sum_{i, j=1}^N \frac{(-1)^{i+j}}{\gcd(i, j) ij}.$$

Write $\hat{v}_{\text{alt}} = v_{\text{alt}}/\|v_{\text{alt}}\|$ and $R(w) = w^\top Q_N w / \|w\|_2^2$.

2 What fails (killed)

Proposition 2.1 (Archived Lemma A is false). *The decomposition $1/\gcd(i, j) = \sum_{d|\gcd(i, j)} \mu(d)\varphi(d)/d^2$ fails already at $\gcd = 2$ ($1/2$ vs. $3/4$). The induced matrix expansion is not Q_N .*

Proposition 2.2 (Gap A for v_{alt} is false as stated). *For $N = 500, 1000$, numerics give $R(v_{\text{alt}}) \approx -0.64, -0.73$ while $-1/(2\pi) \approx -0.159$; the ratio grows with N . Similarly $v_{\text{alt}}^\top Q_N v_{\text{alt}}$ is $\sim 5\times$ more negative than $-\log N/(2\pi)$ and diverges in ratio. Certificate: `scripts/route_c_gap_a_verify.py`.*

Remark 2.3. *High overlap $|\langle v_N^*, \hat{v}_{\text{alt}} \rangle| \approx 0.95$ does not force $R(v_{\text{alt}}) \approx \lambda_{\min}$: $\lambda_{\min} < R(v_{\text{alt}})$ always in checks.*

3 Proved: parity split

Lemma 3.1 (Parity split).

$$\sum_{i,j=1}^N \frac{(-1)^{i+j}}{\gcd(i, j) ij} = \sum_{\substack{d \leq N \\ d \text{ odd}}} \frac{T_-(N/d)}{d^3} + \sum_{\substack{d \leq N \\ d \text{ even}}} \frac{T_+(N/d)}{d^3},$$

where

$$T_-(M) = \sum_{\gcd(m, n)=1} \frac{(-1)^{m+n}}{mn}, \quad T_+(M) = \sum_{\gcd(m, n)=1} \frac{1}{mn}.$$

Proof. Write $i = dm, j = dn$ with $\gcd(m, n) = 1$. The contribution is $(-1)^{d(m+n)}/(d^3 mn)$, which equals $(-1)^{m+n}/(d^3 mn)$ when d is odd and $1/(d^3 mn)$ when d is even. $\square$

Lemma 3.2 (Coprime Möbius forms).

$$T_-(M) = \sum_{d=1}^M \frac{\mu(d)}{d^2} \left(\sum_{m \leq M/d} \frac{(-1)^{dm}}{m} \right)^2, \quad T_+(M) = \sum_{d=1}^M \frac{\mu(d)}{d^2} \left(\sum_{m \leq M/d} \frac{1}{m} \right)^2.$$

Proof. Expand $\sum_{\gcd(m, n)=1} f(m)f(n)/(mn)$ via Möbius inversion $\mathbf{1}_{\gcd=1} = \sum_{d|(m, n)} \mu(d)$ and change variables. $\square$

Remark 3.3. *Even- d blocks contribute positively and cancel a large fraction of odd- d negativity (e.g. at $N = 1000$, even $\approx +4.7$, odd ≈ -10.1). This refutes archived heuristics that even blocks alone drive $-1/(2\pi)$.*

4 Architecture (conditional, reframed)

Theorem 4.1 (Route C template, conditional). *Assume Gap A' (spectral limit $\lambda_{\min}(Q_N)/\log N \rightarrow -1/(2\pi)$ with a rigorous trial-vector lower bound) and Gap B (uniform $\lambda_2(Q_N) - \lambda_{\min}(Q_N) \geq c_0 > 0$). Then Davis–Kahan perturbation yields $\|v_N^* - \hat{v}_{\text{alt}}\| \ll N^{-\eta}$ for some $\eta > 0$.*

Remark 4.2. *The old Gap A target $R(v_{\text{alt}}) \rightarrow -1/(2\pi)$ must not be used: Proposition 2.2. A bridge to zero-density / RH criteria remains separate even if Gaps A'–B close.*

Analytic gaps (open). 1. **Gap A' (spectral).** *Prove $\lim_{N \rightarrow \infty} \frac{\lambda_{\min}(Q_N)}{\log N} = -\frac{1}{2\pi}$ via a trial vector with provable Rayleigh $\sim -(\log N)/(2\pi)$, not v_{alt} as stated in archived drafts.*

2. **Gap B (uniform gap).** *Prove $\lambda_2(Q_N) - \lambda_{\min}(Q_N) \geq c_0 > 0$ uniformly in N (numeric: ≈ 0.09 – 0.14 and slowly increasing through $N = 5000$).*

5 Partial progress on growth (optional)

Using Lemmas 3.1–3.2, one can aim for

$$v_{\text{alt}}^\top Q_N v_{\text{alt}} = -c_1(\log N)^2 + O(\log N)$$

with explicit $c_1 > 0$ from even/odd Möbius blocks. This closes the *growth order* but not the $-1/(2\pi)$ constant for v_{alt} .

6 What is not claimed

- RH or unconditional zero-density theorems.
- $R(v_{\text{alt}}) \rightarrow -1/(2\pi)$ or archived Lemma A.
- Full-spectrum Bridge floors $\lambda_{\min}(Q_N) > -1/2$ (false).

Reproducibility

```
python3 scripts/route_c_gap_a_verify.py 500 1000
python3 scripts/bridge_floor_verify.py 200
```

References

- [1] E. Bombieri, *On the large sieve*, Mathematika **12** (1965), 201–225.
- [2] J.R. Simons, *Route C* (archived draft), Zenodo 10.5281/zenodo.20518388.